

X & Z sono indipendenti
 $p_{X|Z}(x/z) = p_X(x)$

$$p_{X|Y,Z}(x/y,z) = p_{X|Y}(x/y)$$

$$X \rightarrow Y \rightarrow Z$$

$$E[g(X,Y)] = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} g(x,y) p_{X,Y}(x,y) = \sum_{y \in \mathcal{Y}} p_Y(y) \underbrace{\sum_{x \in \mathcal{X}} g(x,y) p_{X|Y}(x/y)}_{h(y)}$$

$$\sum_{x \in \mathcal{X}} g(x,y) \underbrace{p_{X|Y}(x/y)}_{h(y)} = E[g(X,Y) | Y=y] = h(y)$$

$$E[g(X,Y)] = \sum_{y \in \mathcal{Y}} h(y) p_Y(y) = E_Y[h(Y)]$$

$$h(y) = E[g(X,Y) | Y] \implies$$

$$E[g(X,Y)] = E_Y[E[g(X,Y) | Y]]$$

$$E[af(X) + bg(Y)] = aE[f(X)] + bE[g(Y)]$$

$$E[af(X) + bg(Y)] = \sum_{x,y} [af(x) + bg(y)] p_{X,Y}(x,y) =$$

$$= a \sum_{x,y} f(x) p_{X,Y}(x,y) + b \sum_{x,y} g(y) p_{X,Y}(x,y) =$$

$$= a \sum_x f(x) \underbrace{\sum_y p_{X,Y}(x,y)}_{p_X(x)} + b \sum_y g(y) \underbrace{\sum_x p_{X,Y}(x,y)}_{p_Y(y)} =$$

$$= aE[f(X)] + bE[g(Y)]$$

$N \rightarrow \#$ pacchetti in Code o nei router

$N \rightarrow \# \text{ pacchetti in coda a una vocale}$

$$N \in \mathbb{N}_0$$

$$p_N(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

Una funzione p di questi pacchetti ma in una

$M \rightarrow \# \text{ pacchetti in una}$

$$M \in \mathbb{N}_0$$

$$p_M(m) = \sum_{n=0}^{\infty} p_{M|N}(m|n) = \sum_{n=0}^{\infty} p_{M|N}(m|n) p_N(n) = E_N [p_{M|N}^{(m|n)}]$$

$$m \in \mathbb{N}_0$$

$$p_{M|N}(m|n) = \begin{cases} 0, & m > n \\ \binom{n}{m} p^m (1-p)^{n-m}, & m \leq n \end{cases}$$

$$p_M(m) = \sum_{n=m}^{\infty} \underbrace{\frac{n!}{m! (n-m)!}}_{p_{M|N}(m|n)} p^m (1-p)^{n-m} \cdot \underbrace{\frac{\lambda^n}{n!} e^{-\lambda}}_{p_N(n)} =$$

$$= e^{-\lambda} \frac{(\lambda p)^m}{m!} \sum_{n=m}^{\infty} \frac{(1-p)^{n-m}}{(n-m)!} \lambda^{n-m} = e^{-\lambda} \frac{(\lambda p)^m}{m!} \sum_{l=0}^{\infty} \frac{[(1-p)\lambda]^l}{l!}$$

$$e^x = \sum \frac{x^n}{n!} \quad \underbrace{\quad}_{e^{\lambda(1-p)}}$$

$$p_M(m) = e^{-\lambda} e^{\lambda(1-p)} \frac{(\lambda p)^m}{m!} = \frac{(\lambda p)^m}{m!} e^{-\lambda p}$$

(X, Y)

$$\text{cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \underset{g(X, Y)}{=}$$

$$= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} (x - \mu_X)(y - \mu_Y) p_{X, Y}(x, y) = \rho$$

$$\text{Cov}(X, Y) = E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y] =$$

$$= E[XY] - E[X\mu_Y] - E[Y\mu_X] + \mu_X\mu_Y = E[XY] - \mu_X\mu_Y$$

Se sono independente $\Rightarrow$

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

$$\sum_{x,y} (x-\mu_X)(y-\mu_Y) p_X(x)p_Y(y) = \sum_x (x-\mu_X)p_X(x) \sum_y (y-\mu_Y)p_Y(y) =$$

$$= E[X-\mu_X]E[Y-\mu_Y] = 0$$

$$\rho_{X,Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y} \quad -1 \leq \rho_{X,Y} \leq 1$$

$$Z_{\pm} = \frac{X-\mu_X}{\sigma_X} \pm \frac{Y-\mu_Y}{\sigma_Y} \quad E[Z_{\pm}^2] \geq 0$$

$$E[Z_{\pm}^2] = E\left[\left(\frac{X-\mu_X}{\sigma_X} \pm \frac{Y-\mu_Y}{\sigma_Y}\right)^2\right] = E\left[\frac{(X-\mu_X)^2}{\sigma_X^2} + \frac{(Y-\mu_Y)^2}{\sigma_Y^2} \pm$$

$$\pm 2 \frac{(X-\mu_X)(Y-\mu_Y)}{\sigma_X \sigma_Y}\right] = E\left[\frac{(X-\mu_X)^2}{\sigma_X^2}\right] + E\left[\frac{(Y-\mu_Y)^2}{\sigma_Y^2}\right] \pm$$

$$\pm 2 E\left[\frac{(X-\mu_X)(Y-\mu_Y)}{\sigma_X \sigma_Y}\right] = 1 + 1 \pm 2\rho_{X,Y} \Rightarrow$$

$$2(1 \pm \rho_{X,Y}) \geq 0 \quad -1 \leq \rho_{X,Y} \leq 1$$

$$E[(X-\mu_X)(Y-\mu_Y)] \quad E[(X-\mu_X)^2] = \sigma_X^2$$

$$\langle X, Y \rangle = \text{Cov}(X, Y) \quad \|X\| = \sigma_X \quad \|Y\| = \sigma_Y$$

$$|\langle X, Y \rangle| \leq \|X\| \|Y\| \Rightarrow |\text{Cov}(X, Y)| \leq \sigma_X \sigma_Y$$

$$|\rho_{X,Y}| \leq 1 \Rightarrow -1 \leq \rho_{X,Y} \leq 1$$

$$\rho_{X,Y} = 1$$

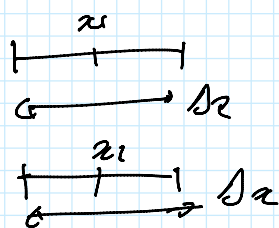
$$(X, Y)$$

$$(X, Y)$$

$$\begin{aligned}
 & \rho_{X,Y} = 1 \\
 & \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \underline{X} \\
 & (X_1, X_2) \rightarrow \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \\
 & \underline{X} = \underline{X}(\omega) \quad E[\underline{X}] = \begin{pmatrix} E[X_1] \\ E[X_2] \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \\
 & = \underline{\mu} \in \mathbb{R}^{2 \times 1}
 \end{aligned}$$

$$\begin{aligned}
 & \underbrace{\begin{pmatrix} \underline{X} - \underline{\mu} \end{pmatrix}}_{\mathbb{R}^{2 \times 1}} \underbrace{\begin{pmatrix} \underline{X} - \underline{\mu} \end{pmatrix}^T}_{\mathbb{R}^{1 \times 2}} = \begin{pmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{pmatrix} \underbrace{\begin{pmatrix} X_1 - \mu_1 & X_2 - \mu_2 \end{pmatrix}}_{\mathbb{R}^{2 \times 2}} = \\
 & = \begin{pmatrix} (X_1 - \mu_1)^2 & (X_1 - \mu_1)(X_2 - \mu_2) \\ (X_1 - \mu_1)(X_2 - \mu_2) & (X_2 - \mu_2)^2 \end{pmatrix} \\
 & \underline{K} = E \left[\begin{pmatrix} \underline{X} - \underline{\mu} \end{pmatrix} \begin{pmatrix} \underline{X} - \underline{\mu} \end{pmatrix}^T \right] = \begin{pmatrix} E[(X_1 - \mu_1)^2] & E[(X_1 - \mu_1)(X_2 - \mu_2)] \\ E[(X_1 - \mu_1)(X_2 - \mu_2)] & E[(X_2 - \mu_2)^2] \end{pmatrix} = \\
 & = \begin{pmatrix} \sigma_1^2 & \text{cov}(X_1, X_2) \\ \text{cov}(X_1, X_2) & \sigma_2^2 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 & (\Omega, \mathcal{E}, \mathbb{P}) \\
 & X: \omega \in \Omega \rightarrow X(\omega) \in \mathcal{X} \subseteq \mathbb{R} \\
 & \left\{ a - \frac{\Delta x}{2} \leq X \leq a + \frac{\Delta x}{2} \right\}
 \end{aligned}$$



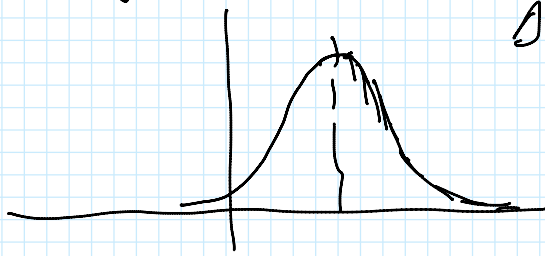
$$\begin{aligned}
 & \Delta x \\
 & M(x_1; \Delta x) \\
 & M(x_2; \Delta x)
 \end{aligned}$$

$$\begin{aligned}
 & \geq 0 \\
 & A \cap B = \emptyset \\
 & \mu(A \cup B) = \mu(A) + \mu(B)
 \end{aligned}$$

$$\rho(x_1) = \lim_{\Delta x \rightarrow 0} \frac{M(x_1; \Delta x)}{\Delta x}$$

$$\rho \mid_{a-\Delta x} \mid_{a+\Delta x} \quad \mathbb{P} \left\{ a - \Delta x \leq X \leq a + \Delta x \right\}$$

$$f_X(x_1) = \lim_{\Delta x \rightarrow 0} \frac{P\left\{x_1 - \frac{\Delta x}{2} \leq X \leq x_1 + \frac{\Delta x}{2}\right\}}{\Delta x}$$



$$P\left\{X \in \left(x_1 - \frac{\Delta x}{2}, x_1 + \frac{\Delta x}{2}\right)\right\} \stackrel{\Delta x \rightarrow 0}{\approx} f_X(x_1) \Delta x$$